Tsinghua University 2024 Student Academic Research Promotion Program
Individual Project Mid-term Exchange

Phage therapy targeting *Eggerthella lenta* for depression

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1. Background

Current treatment for Major Depressive Disorder(MDD):

- About 75% patients without medical treatment;
- Most available medicine provokes strong side effects such as weight gain and drowsiness;

MDD and gut-brain axis:

- Gut microbiome transplantation from MDD patients to germ-free mice can induce MDD;
- Rising level of endogenous *Eggerthella lenta* is prominently associated to MDD

Phage therapy for MDD:

- Phages therapy has no side effects!
- No lytic phage that reduces endogenous *E. lenta* level is found yet

Alleviate MDD←reduce endogenous *E. lenta* level←phage against *E. lenta*

2. Methods

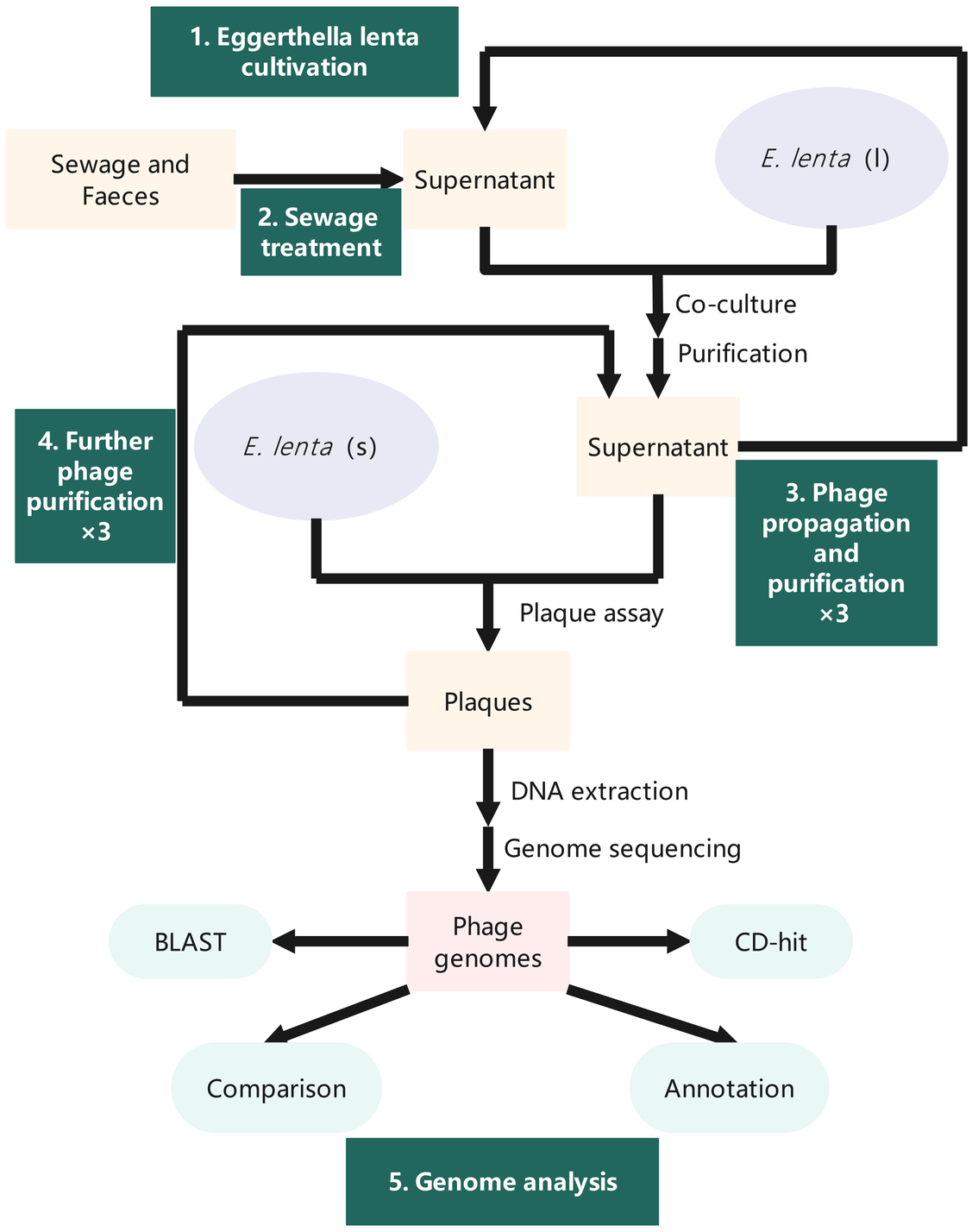


Figure 1 Flowchart of experimental procedure

3. Results - Isolation of 2 novel phages against *E.lenta* DSM2243 (PhiEL2-3 and PhiEL3-3)

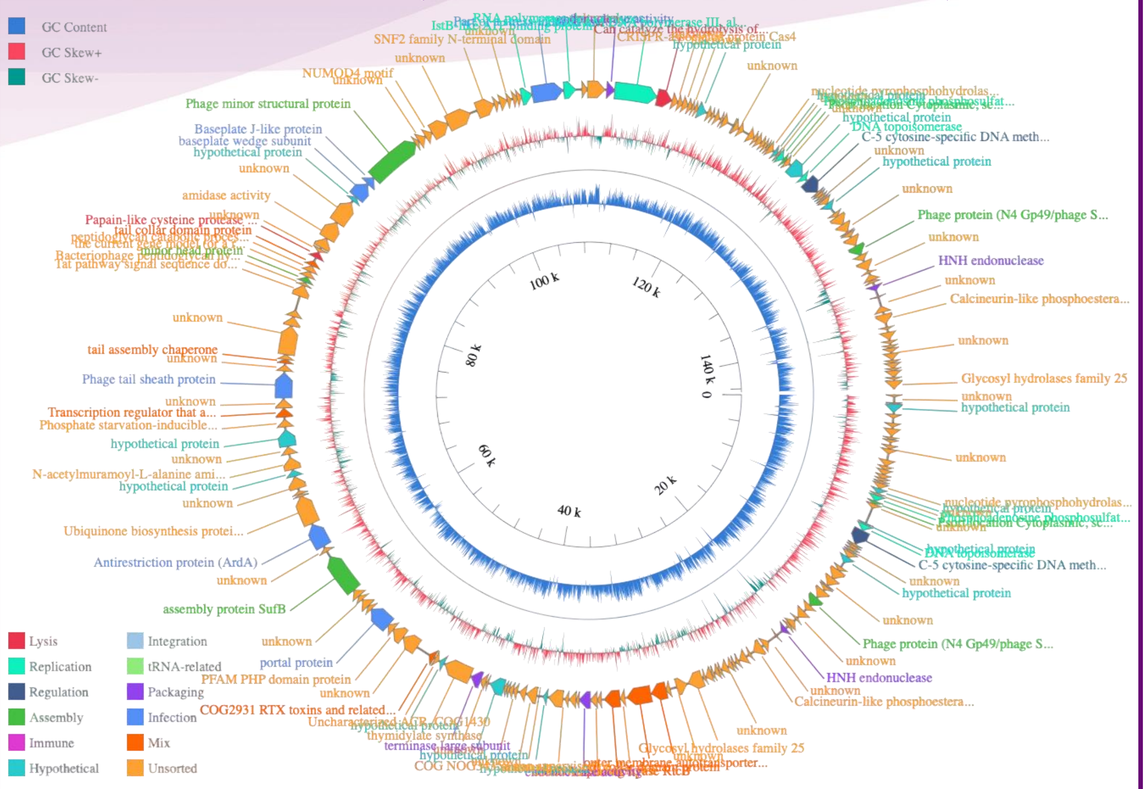


Figure 2 Annotated genome map & overview of PhiEL2-3



- Both virulent, similarly annotated
- BLAST result: similar to LE2-3（Per. Ident 98.89%, E value 0)
- Rich in genes concerning nucleic acid metabolism
- CRISPR-assiciated protein Cas4 in both phages and in *E. lenta*
- Hypothesis on Cas4: retard bacterial resistance as RNAi

4. Future Research Plan

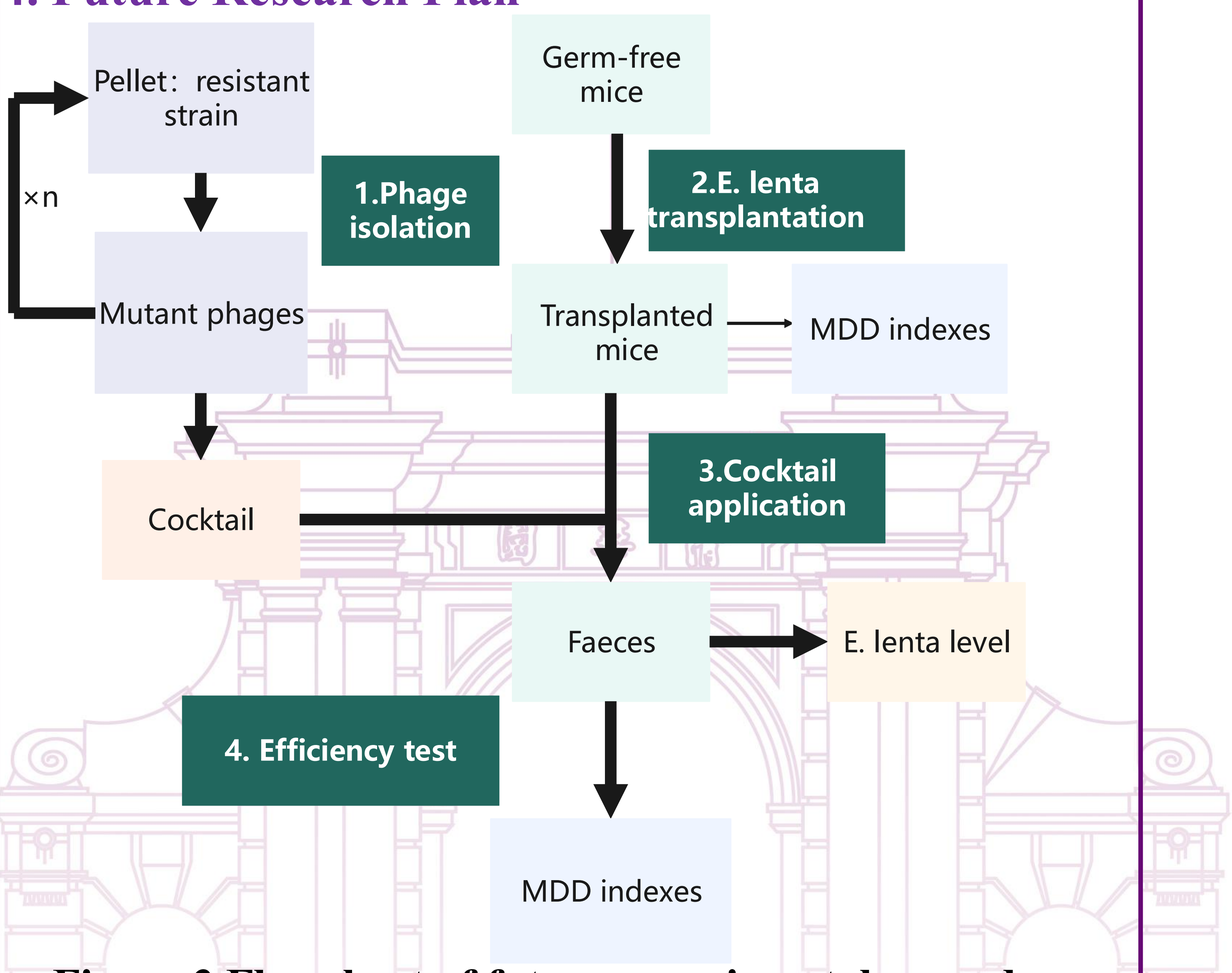


Figure 3 Flowchart of future experimental procedure

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